

Beta Calibration

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Motivation

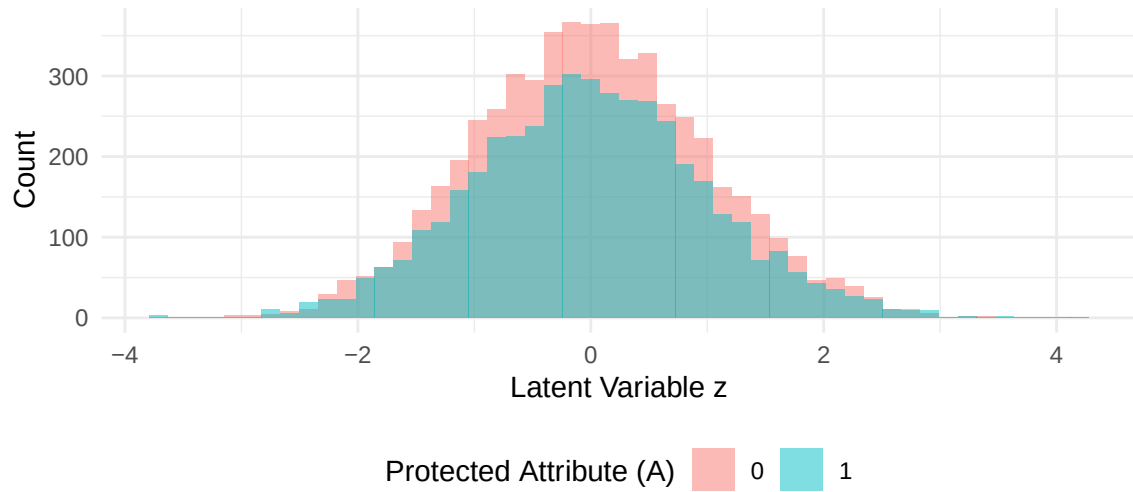
Base Case

Latent Variable Generation

Generate latent variable Z as

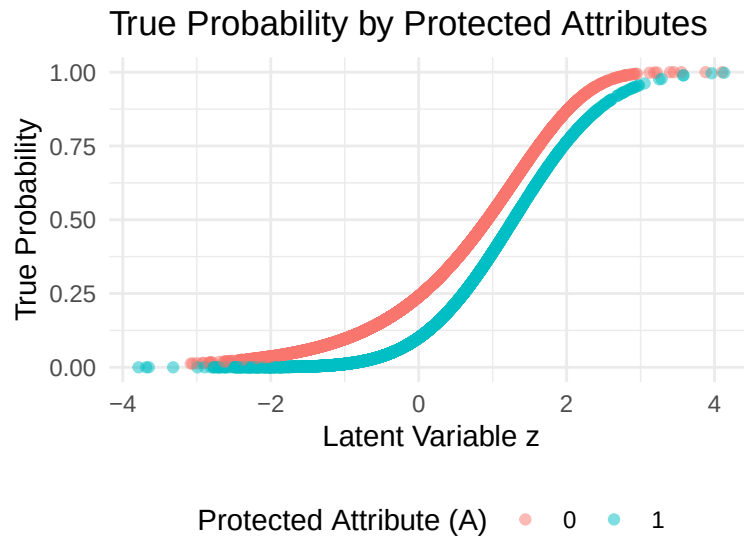
$$Z_a = \beta_a^T \mathbf{X} + \eta_a^T \mathbf{W} + \text{interactions} + \text{nonlinear terms}$$

Histogram of z by Protected Attribute A



Inverse cloglog link

$$P(Y = 1|Z_a) = 1 - \exp(-\exp(Z_a))$$

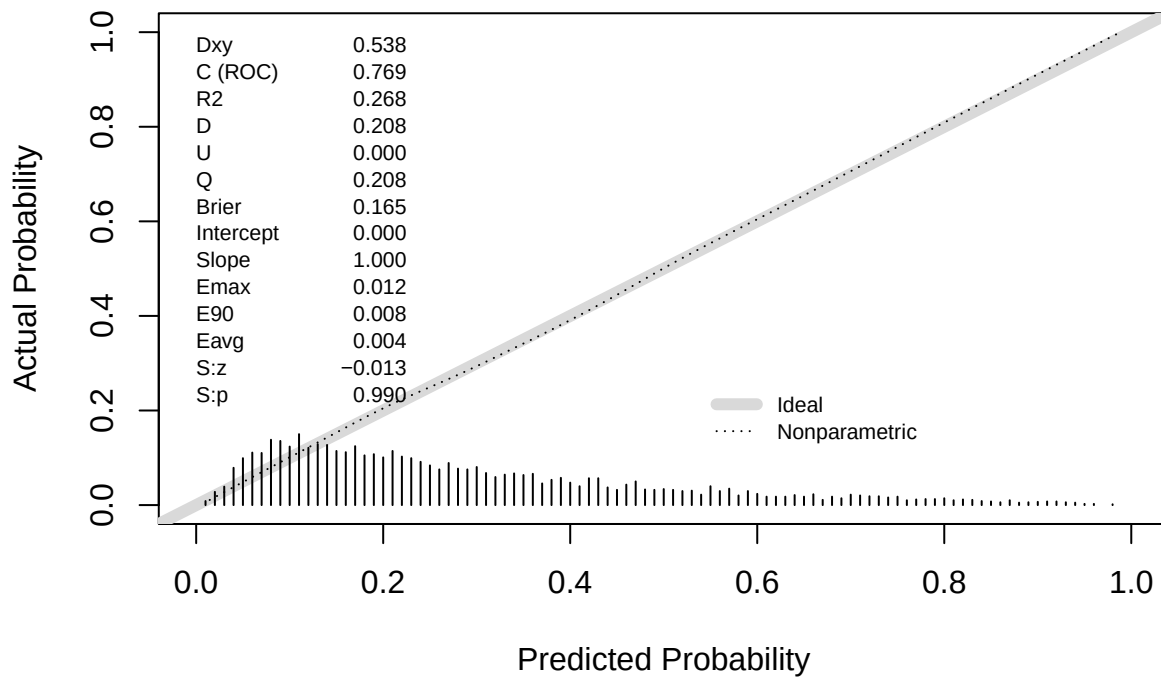


Model Fitting and Calibration

Regress $Y \sim X$ only within each group $A = 0$ and $A = 1$ separately. Then calibrate the predicted probabilities using Beta Calibration within each group.

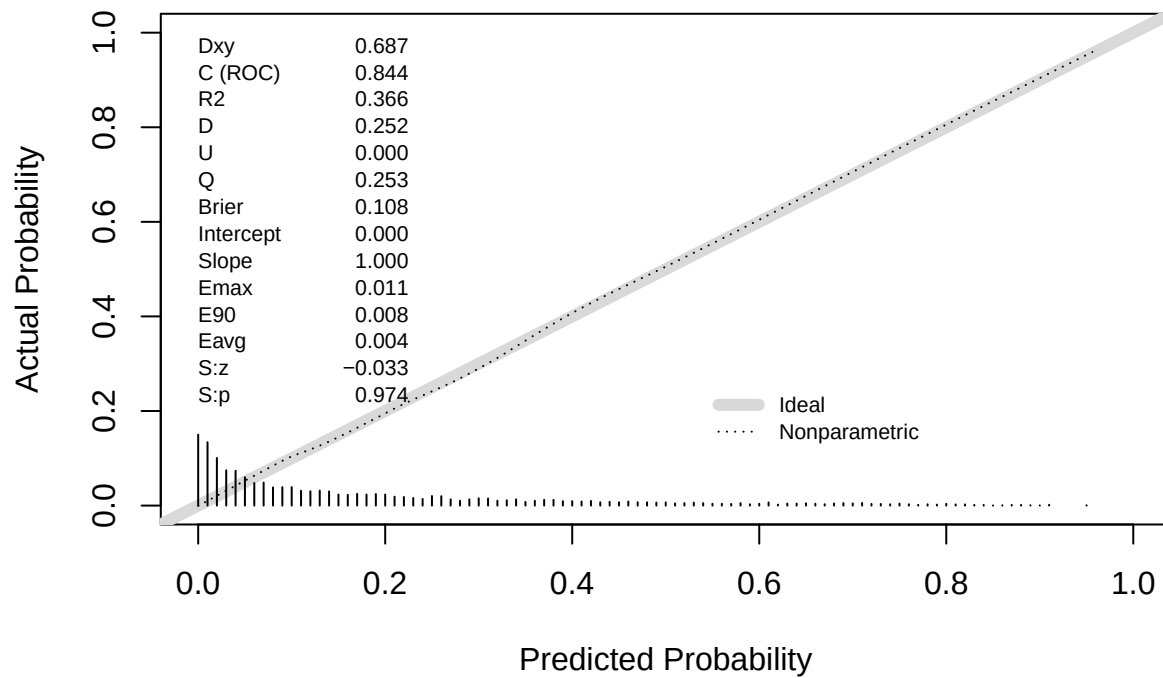
```
## [1] -25.96054
```

```
## [1] 57.63025
```



```
## [1] -42.01642
```

```
## [1] 25.07899
```



```
## Warning in as_grob.default(plot): Cannot convert object of class numeric into a
## grob.
## Warning in as_grob.default(plot): Cannot convert object of class numeric into a
## grob.
```

Modified Case:

Add another layer of nonlinearity to $g_a(z)$ to create mirror-opposite groups

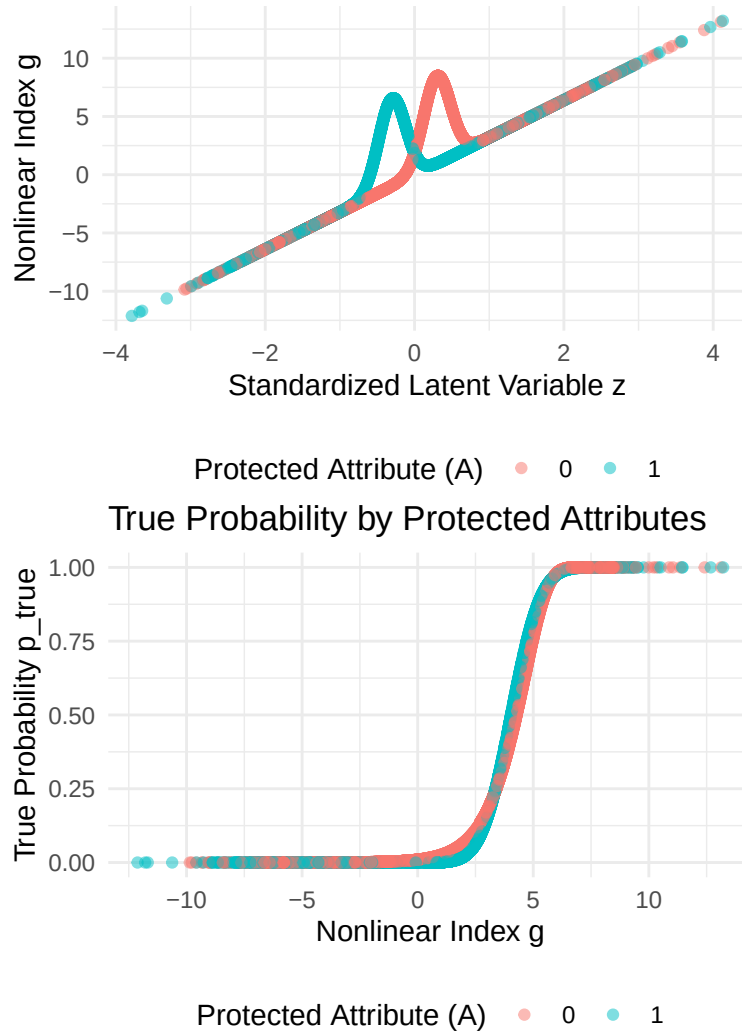
Define

$$g_0(z) = 3.2 * z + 7.5 * \exp(-(z - 0.3)^2 / (2 * 0.18^2))$$

and

$$g_1(z) = 3.2 * z + 7.5 * \exp(-(z + 0.3)^2 / (2 * 0.18^2))$$

The shift in groups is needed, otherwise the bias will be in the same direction for both groups, and canceled out for delta. Then use same link functions as before to get p_true .

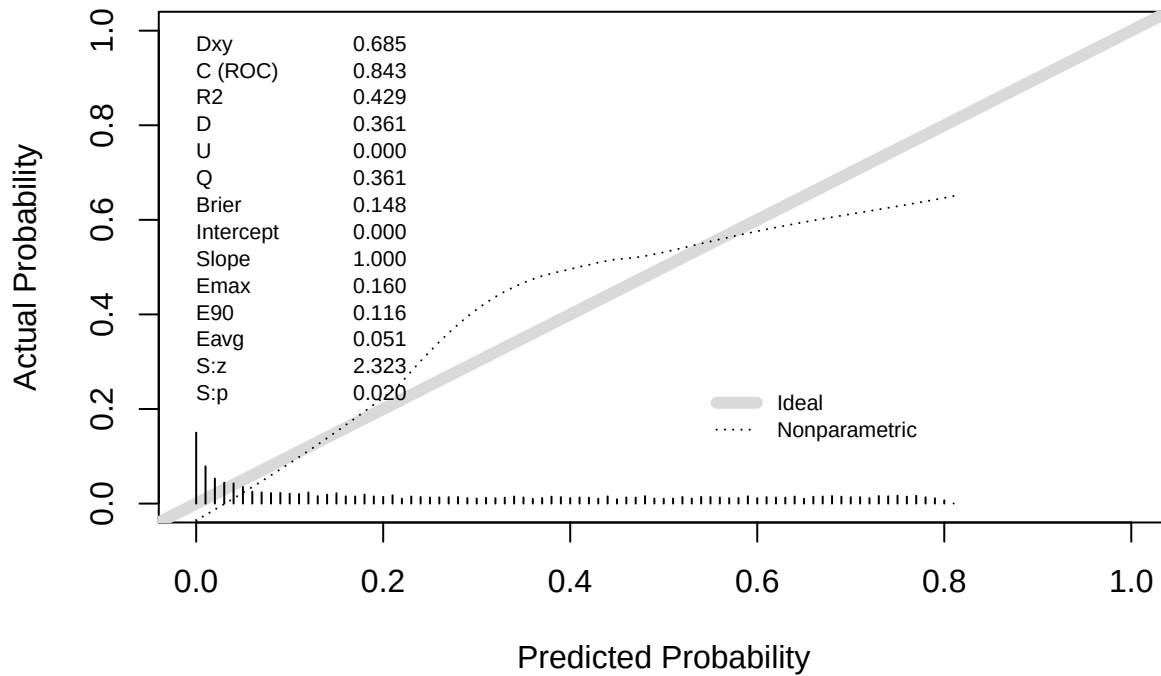


Model Fitting and Calibration

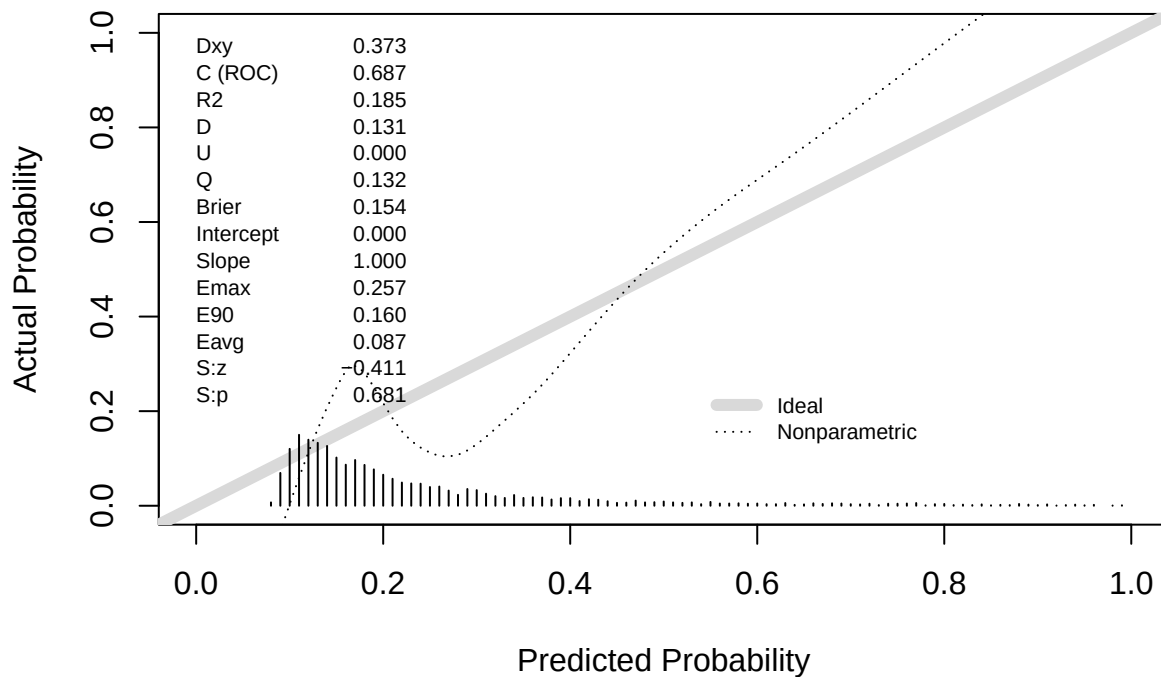
Regress $Y \sim X$ only within each group $A = 0$ and $A = 1$ separately. Then calibrate the predicted probabilities using Beta Calibration within each group.

```
## [1] -66.65601
```

```
## [1] -1.467032
```

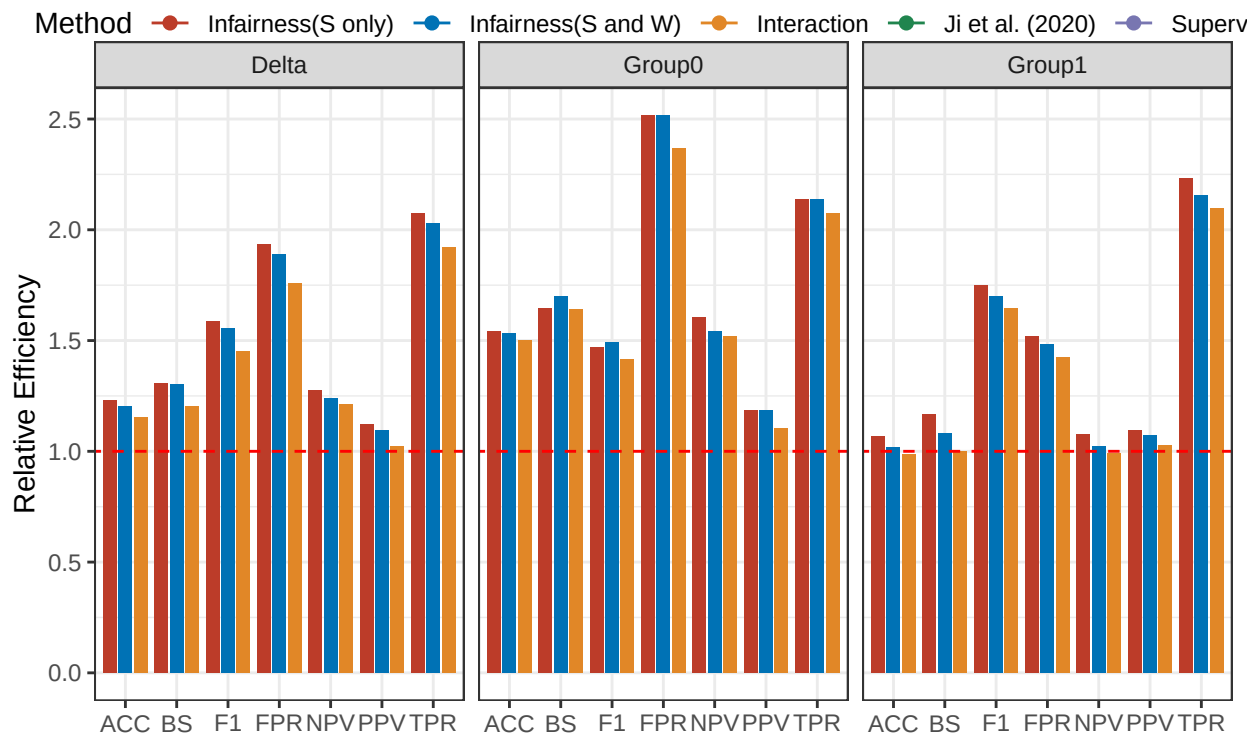
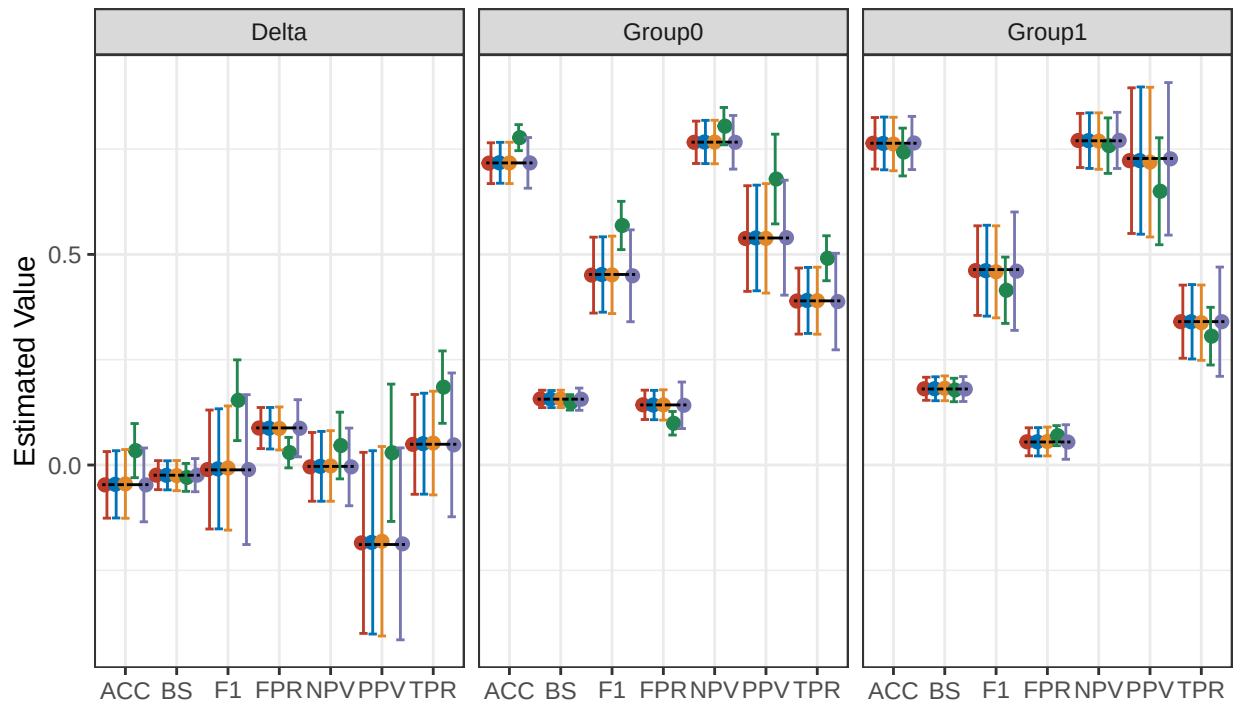


```
##          Dxy          C (ROC)          R2          D          D:Chi-sq
## 6.852085e-01 8.426043e-01 4.292742e-01 3.610647e-01 2.004548e+03
##          D:p          U          U:Chi-sq          U:p          Q
## 0.000000e+00 -3.604253e-04 -1.546141e-11 1.000000e+00 3.614252e-01
##          Brier          Intercept          Slope          Emax          E90
## 1.483238e-01 1.792068e-13 1.000000e+00 1.600658e-01 1.156056e-01
##          Eavg          S:z          S:p
## 5.119316e-02 2.323115e+00 2.017298e-02
## [1] 1.522292
## [1] 135.9375
```

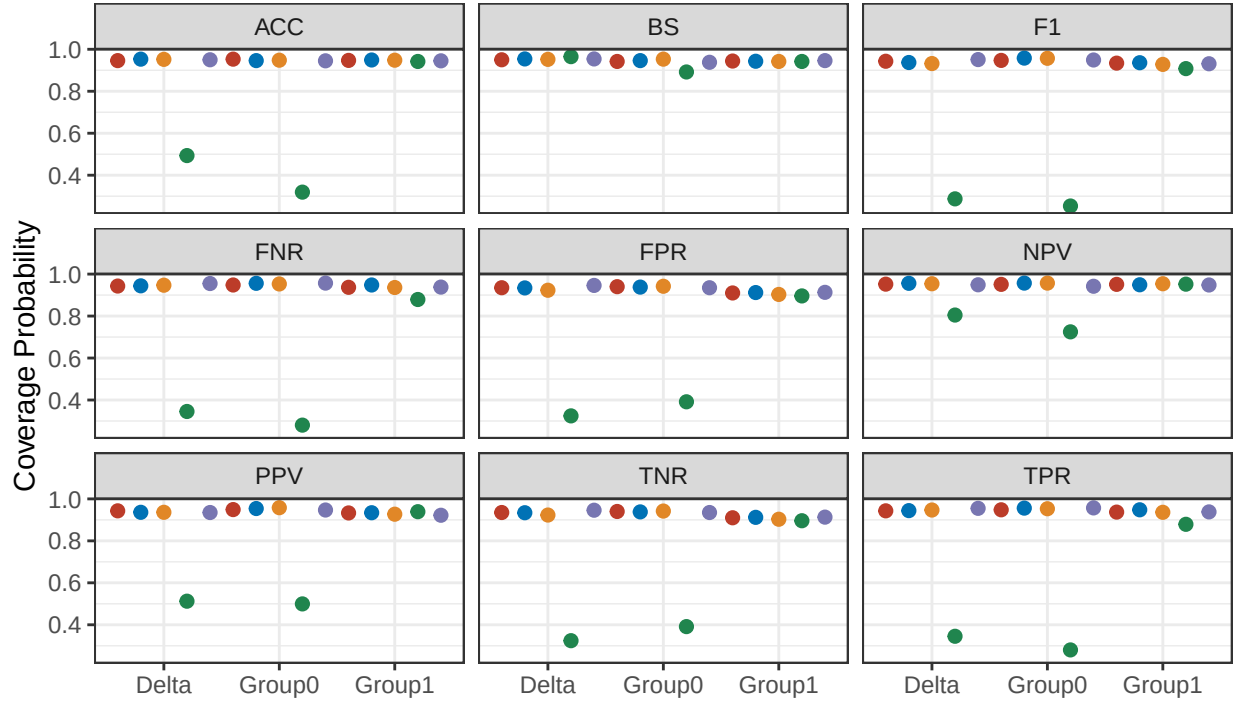


##	Dxy	C (ROC)	R2	D	D:Chi-sq
##	3.730029e-01	6.865014e-01	1.848326e-01	1.314804e-01	5.862191e+02
##	D:p	U	U:Chi-sq	U:p	Q
##	0.000000e+00	-4.493372e-04	-1.364242e-11	1.000000e+00	1.319297e-01
##	Brier	Intercept	Slope	E _{max}	E ₉₀
##	1.543890e-01	2.480845e-10	1.000000e+00	2.567439e-01	1.603941e-01
##	E _{avg}	S:z	S:p		
##	8.678684e-02	-4.112154e-01	6.809146e-01		

Results



Method Infairness(S only) Infairness(S and W) Interaction



Method ● Infairness(S only) ● Infairness(S and W) ● Interaction ● Ji et al. (2020) ● Superv

##	Method	Group	Brier_Score	Log_Loss	AUC	ECE
## 1	ss_s	g0	0.1405438	0.4255406	0.8506943	0.06794898
## 2	ss_s	g1	0.1648062	0.5285802	0.7517040	0.08348616
## 3	ss_sw	g0	0.1459213	0.4397635	0.8489259	0.07942175
## 4	ss_sw	g1	0.1733553	0.5479101	0.7266966	0.08283576
## 5	ss_sw_int	g0	0.1509939	0.4584300	0.8408587	0.08465468
## 6	ss_sw_int	g1	0.1803870	0.5667960	0.6993746	0.09567766